# AFFORD-Bench: An Affordance-Grounded Benchmark and Generalization Evaluation for Robot Learning

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***Abstract*—Imitation learning policies and Vision-Language-Action (VLA) models report high success on standard manipulation benchmarks, yet LIBERO-PRO and LIBERO-X show that these numbers collapse under modest changes to object layout, appearance, or instruction wording, which points to memorization of nominal scenes. Existing robustness suites perturb mostly tabletop pick-and-place scenes and rarely vary the quantities that set whether a contact-rich task can succeed: placement tolerance, friction, and mass. We introduce AFFORD-Bench, a policy-agnostic MuJoCo benchmark with 210 tasks across 7 domains (industrial assembly, culinary, domestic tabletop, laboratory, long-horizon, precision chains, and service chains) on a simulated Franka Emika Panda. It ships oracle demonstrations with dual-camera RGB ($640 \times 660$ workcell, $300 \times 420$ wrist), optional metric depth, and proprioception in Robomimic HDF5, together with four instruction variants per task (metric, contact-conditioned, terse, and subtask-decomposed). We specify a five-level perturbation hierarchy (L1–L5) over spatial, photometric, physical, and compound shifts, organized as 35 evaluation suites, all implemented. Grasping is frictional, and success is a physical predicate on the settled state: the released body must rest within 2–2.5 mm and $5^\circ$ of its goal in the precision domains (8 mm and $15^\circ$ elsewhere). A closed-loop privileged oracle solves 99.8% of nominal episodes, while null and uniform-random policies solve none.**

## I. Introduction

Generalist Vision-Language-Action (VLA) models [10, 15, 16], continuous flow models such as $\pi_0$ [1], 3D spatial architectures such as SpatialVLA [18], visual reasoning policies such as CoT-VLA [28], compact VLAs such as SmolVLA [20], subtask-aware post-training such as AtomVLA [22], and action-chunking visuomotor policies [5, 27] all promise transfer across tasks and scenes.

Measuring that transfer remains difficult. LIBERO [12] is the de facto testbed, but Zhou et al. [29] (LIBERO-PRO) and Wang et al. [25] (LIBERO-X) show that policies exceeding 90% nominal success can fall to near 0% under small spatial offsets, object swaps, or altered instructions, and attribute the gap to memorized trajectories and layouts.

LIBERO-PRO and LIBERO-X perturb layout, appearance, and language; COLOSSEUM [17] adds physical-property axes over 20 RLBench tasks. None of them couples these shifts to the quantity that decides success in precision work: how much positional error the task tolerates. In the sense of affordance theory [7], whether a part can be seated depends jointly on clearance, friction, and mass, not on appearance alone. Contact-rich assembly simulators such as Factory [14] and IndustReal [23] model these effects with high fidelity but target reinforcement learning on a few assemblies, not cross-domain evaluation of imitation and VLA policies.

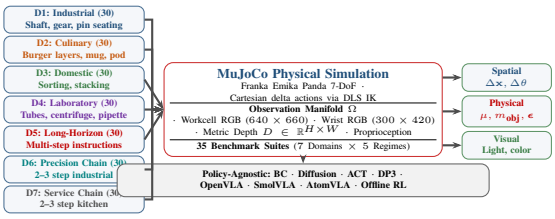



Fig. 1. **The AFFORD-Bench Architecture.** 210 tasks in 7 domains, evaluated under spatial, physical, and visual shifts in 35 suites.

A benchmark should also be **policy-agnostic**: usable by behavior cloning, chunked diffusion and transformer policies (Diffusion Policy, ACT), point-cloud policies (DP3), VLAs (OpenVLA, SmolVLA, AtomVLA), and offline reinforcement learning (CQL, IQL).

AFFORD-Bench contributes:

1) **Task scale and domain spread**: **210 tasks** in **7 domains**, spanning industrial seating, culinary stacking, tabletop sorting, laboratory handling, and multi-step instructions.
2) **Policy-agnostic data**: dual-camera RGB ($640 \times 660$ workcell, $300 \times 420$ wrist), optional metric depth, joint and Cartesian proprioception, and success flags in Robomimic HDF5 [13], convertible to RLDS and LeRobot formats.
3) **Physical success at millimeter tolerance**: frictional, force-limited grasping with no attachment or goal snapping; success requires the released body to rest, supported, within 2–2.5 mm and $5^\circ$ of its goal in the precision domains (8 mm and $15^\circ$ elsewhere).
4) **Hierarchical perturbations**: a five-level hierarchy (L1 nominal, L2 spatial, L3 visual, L4 physical, L5 compound) yielding **35 suites**, all implemented (Sec. IV).
5) **Multi-tier language**: four instruction variants per task (metric, contact-conditioned, terse, and subtask-decomposed with pre/post predicates), each annotated with whether the simulator evaluates what it asks.

## II. Related Work

### A. Robotic Manipulation Benchmarks

RLBench [9] and ManiSkill2 [8] provide simulated suites for reinforcement and demonstration-guided learning. Robomimic [13] standardized offline imitation learning

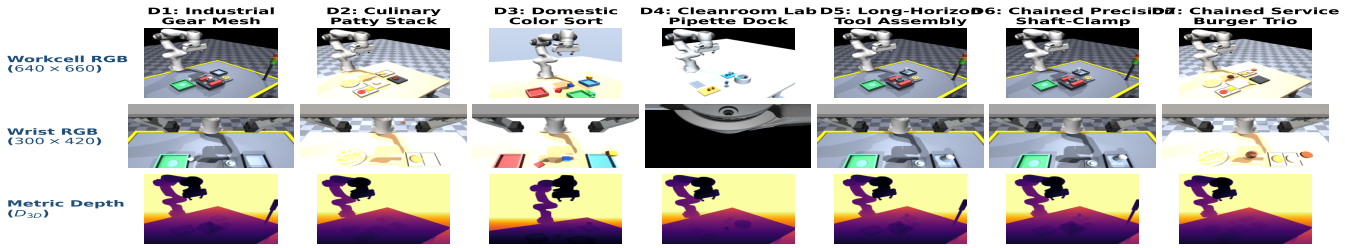



Fig. 2. **Multi-Domain Qualitative Visualizations in AFFORD-Bench.** Representative tasks from all seven domains: (Row 1) workcell RGB (640 × 660); (Row 2) wrist RGB (300 × 420); (Row 3) metric depth rendered from the workcell camera (meters).

pipelines across demonstration qualities. LIBERO [12] introduced 130 tasks for lifelong knowledge transfer. These suites evaluate mostly nominal execution without systematic out-of-distribution (OOD) testing.

### B. OOD Robustness and Modern VLA Architectures

With large-scale pretrained policies [16, 3, 2, 10, 15], robustness to distribution shift has become a primary question. LIBERO-PRO [29] shows that policies with high nominal LIBERO success collapse under perturbed objects, initial states, and instructions. LIBERO-X [25] adds hierarchical, cumulative perturbation levels. COLOSSEUM [17] varies 14 factors, including physical properties, and reports 30–50% degradation for single factors and at least 75% for combined ones. Flow models such as $\pi_0$ [1] predict continuous actions at high frequency, SpatialVLA [18] embeds 3D representations, CoT-VLA [28] generates visual reasoning traces, SmolVLA [20] pairs a compact backbone with flow matching, and AtomVLA [22] combines subtask decomposition with a predictive world model. These evaluations remain concentrated on domestic tabletop scenes with loose placement tolerances.

### C. Affordance and Contact-Rich Manipulation

Affordance theory [7] frames objects by the interactions they permit. In robotics this has motivated contact-rich grasping [21], spatial affordance maps [19], and 3D policy representations such as DP3 [26]. Factory [14] and IndustReal [23] simulate contact-rich assembly with SDF collisions and transfer RL insertion and gear-meshing policies to a real Franka. Diffusion Policy [5] and ACT [27] reach high precision on low-cost hardware [6], but how they degrade as tolerance tightens or friction changes is not characterized at scale. Offline RL [11] additionally needs transitions and rewards that VLA-specific datasets often omit. AFFORD-Bench targets this combination of tolerance-indexed tasks, multi-domain coverage, and policy-agnostic data.

## III. The AFFORD-Bench Formulation

### A. Tasks as Goal-Conditioned MDPs

Each task is a tuple $\kappa = (\ell, \mathbf{e}, \mathcal{C}_{\text{aff}})$ with instruction $\ell$, environment $\mathbf{e} = (\mathcal{W}, \mathcal{O}, \rho_0, \Phi)$, and goal set $\mathcal{C}_{\text{aff}}$. $\mathcal{W}$ is the workspace geometry, $\mathcal{O} = \{o_1, \ldots, o_K\}$ the object set with one evaluated body $o^\star \in \mathcal{O}$, $\rho_0$ the initial-state distribution, and $\Phi = (\mu, m, \boldsymbol{\epsilon})$ the physical parameters: sliding friction $\mu$ of $o^\star$, its mass $m$, and success tolerance $\boldsymbol{\epsilon} = (\epsilon_{xy}, \epsilon_z)$.

Each task induces an MDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, P, R, \Omega, H \rangle$. $\mathcal{S}$ is the joint robot-object state. An action $\mathbf{a} = (\Delta\mathbf{p}, \Delta\boldsymbol{\theta}, a_{\text{grip}}) \in [-1, 1]^7$ commands a TCP position increment $0.025\,\Delta\mathbf{p}$ m, capped at 2.5 cm per step (0.5 m/s) and tracked by damped-least-squares IK at 20 Hz, a rotation increment $\Delta\boldsymbol{\theta}$ of which only the yaw component is executed ($0.05\,\Delta\theta_z$ rad per step, hand pointing down), and a per-finger opening target $w = 0.02\,(a_{\text{grip}} + 1)$ m, approached at no more than 0.05 m/s by a force-limited parallel jaw (about 35 N per finger). $P(s' \mid s, \mathbf{a})$ is given by MuJoCo [24] and is deterministic given $s$; all stochasticity enters through $\rho_0$ and the perturbation sampler, both seeded per episode. $R(s) = \mathbb{I}[s \in \mathcal{C}_{\text{aff}}]$ is sparse, and $H$ is the step budget.

The goal set is a set of physical states:

$$\begin{aligned}\mathcal{C}_{\text{aff}} = \big\{ s \mid \|\mathbf{p}^{xy}_{o^\star} - \mathbf{g}^{xy}\|_2 \le \epsilon_{xy},\ |p^z_{o^\star} - g^z| \le \epsilon_z,\\ \theta_{o^\star} \le \epsilon_\theta,\ \mathcal{F}(o^\star) = \emptyset,\ \mathcal{K}(o^\star) \neq \emptyset,\\ \|\mathbf{v}_{o^\star}\| < v_{\text{rest}},\ \|\boldsymbol{\omega}_{o^\star}\| < \omega_{\text{rest}} \big\},\end{aligned} \tag{1}$$

where $\theta_{o^\star}$ is the tilt of $o^\star$ from upright, $\mathcal{F}$ and $\mathcal{K}$ are the finger bodies and the non-robot bodies in contact with $o^\star$, $v_{\text{rest}} = 0.02$ m/s, and $\omega_{\text{rest}} = 0.2$ rad/s. The goal $\mathbf{g}$ is the resting pose of $o^\star$ at the task's target, found when the environment is built by lowering $o^\star$ onto its support and simulating it to rest; there is no attachment, snapping, or teleport, so a body that slides, tips, or binds on a rim fails. Targets with no stable resting pose, such as holding a pipette above a stand, are hold goals, for which $\mathcal{F}(o^\star)$ must contain both fingers instead. Every manipulation task also requires $o^\star$ to have been lifted by $h_{\min} = 2$ cm (10–15 cm for four lift-and-replace tasks), so pushing $o^\star$ to the goal does not count. For reach-and-press tasks the condition is $\|\mathbf{p}_{\text{tcp}} - \mathbf{g}\|_2 \le \epsilon_{xy}$ for 5 consecutive steps. Episodes terminate at the first $t \le H$ with $s_t \in \mathcal{C}_{\text{aff}}$, so policy competence is

$$J(\pi; \kappa) = \mathbb{P}_{s_0 \sim \rho_0,\ \tau \sim \pi}\big[\exists\, t \le H : s_t \in \mathcal{C}_{\text{aff}}\big]. \tag{2}$$

Contact forces are monitored, not part of success: an episode violates safety if the net force $\mathbf{f}_c(s_t)$ that the scene, excluding $o^\star$, exerts on the robot exceeds $F_{\max} = 150$ N at some control step. One oracle episode per task peaked at 149 N (median 0 N), so the threshold flags contact harder than the scripted expert's in any task.

### B. Policy-Agnostic Benchmark Architecture

The observation $\mathbf{o}_t \in \Omega$ comprises:

1) **Workcell RGB**: $H \times W = 640 \times 660$, uint8;
2) **Wrist RGB**: $300 \times 420$, uint8, mounted at the gripper;
3) **Metric depth** (optional): float32 $D \in \mathbb{R}^{640\times660}$ in meters, back-projected as $\mathbf{P} = D(u,v)\,\mathbf{K}^{-1}[u,v,1]^\top$ using the camera intrinsics $\mathbf{K}$ and pose stored with each dataset (`generate_demos.py --depth`);
4) **Proprioception**: joint positions $\mathbf{q} \in \mathbb{R}^7$, velocities $\dot{\mathbf{q}} \in \mathbb{R}^7$, TCP pose $\mathbf{x}_{\text{tcp}} \in \mathbb{R}^7$ (position and $w$-first quaternion), and per-finger opening $w \in [0, 0.04]$ m;
5) **Rewards and termination**: per-step sparse reward $R(s_t)$ and success flag.

### C. Domain Architecture

As summarized in Table I and Figure 2, AFFORD-Bench has seven domains of 30 tasks each across five scenes: **D1: Industrial Precision** (001–030): shaft, gear, and pin seating in a jig, kitting, and fixture contact. **D2: Culinary Operations** (031–060): burger layer stacking, mug and capsule handling at an espresso station. **D3: Domestic Tabletop** (061–090): color sorting into bins, stacking, and relocation. **D4: Laboratory Sciences** (091–120): tube racking, centrifuge loading, and pipette positioning. **D5–D7** (121–210): instructions that chain 2–4 steps across industrial, laboratory, and kitchen scenes. In Stage 1 each of these tasks evaluates a single body, and the episode ends when that body reaches its goal, which in 57 of 90 tasks occurs before the last instructed step. Multi-body sequential predicates that score every step are a Stage 2 deliverable.

## IV. The Hierarchical Affordance Perturbation Framework

AFFORD-Bench organizes distribution shift into five levels:

**L1: Nominal (in-distribution).** The start of $o^\star$ is drawn from $\mathcal{U}(\pm 7\,\text{cm})$ per horizontal axis around its nominal position, yaw from $\mathcal{U}(-45°, 45°)$, and roll and pitch from $\mathcal{U}(-10°, 10°)$ at spawn, resolved by simulating the body to rest; the initial TCP is jittered by $\pm 1.5$ cm per horizontal axis. Starts are rejection-sampled: $o^\star$ must rest flat on a single support (tilt below 3°, support height spread at most 2 mm), have no fixture overhead, admit a collision-free top-down parallel-jaw grasp, and lie away from the goal. If no valid start is found, the scatter is shrunk to $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{1}{10}$ before falling back to the nominal pose, so bodies that start in a compartment or seated on a shaft vary less than the nominal range. D1, D3, and D7 scenes add two distractor bodies of similar color or shape, dropped at random workspace positions at least 9 cm from the start and goal.

**L2: Spatial ($\mathcal{P}_{\text{spatial}}$).** The start offset has radius $\mathcal{U}(7, 12)$ cm in a uniformly random direction, outside the L1 support, and yaw magnitude $\mathcal{U}(45°, 90°)$, probing closed-loop servoing versus trajectory recall.

**L3: Visual ($\mathcal{P}_{\text{visual}}$).** Light intensity in $[0.5, 1.5]\,I_0$, light position jitter of $\pm 0.5$ m, and object color jitter of $\pm 25\%$, leaving geometry unchanged.

**L4: Physical ($\mathcal{P}_{\text{physics}}$).** Sliding friction $\mu \sim \mathcal{U}(0.12, 0.25)$ on $o^\star$ and on both finger pads, and mass and rotational inertia of $o^\star$ scaled by $\mathcal{U}(1.8, 2.5)$, at identical visual coordinates. MuJoCo takes the larger friction of a contact pair, so pads and body are set together. In a probe with the oracle's grasp shifted 10–20 mm along the hand's lateral axis, lifted 10 cm, and carried 20 cm at the speed cap (3 seeds per setting), L4 dropped 8 of 12 off-center grasps of a cube and a mug, against 0 of 12 at nominal; centered grasps of the same bodies were held but slid 3–13 mm, and the 24 mm shaft was dropped at L4 even when grasped centrally.

**L5: Compound ($\mathcal{P}_{\text{comp}}$).** L2, L3, L4, and color shifts sampled jointly, combined with the multi-step instructions of D5–D7.

**Physics and implementation.** All five levels are implemented. Lights, colors, friction, and masses are restored to nominal before each sample, and all sampling is seeded per episode. Grasping is frictional: the arm is driven by gravity-compensated joint position servos, the gripper is a force-limited tendon actuator, finger pads use stiff contacts (4 ms time constant), and collisions use MuJoCo's native convex collision with multi-point contact. At reset, bodies are lifted out of any interpenetration and simulated to rest before the first observation.

## V. Algorithmic Oracles and Dataset Synthesis

### A. Closed-Loop Oracles

Teleoperated demonstrations are costly and inconsistent [13]. AFFORD-Bench instead uses a single privileged, closed-loop oracle for all 210 tasks that reads the pose of $o^\star$ at every step and acts through the same controller and action space as learned policies. It plans a grasp by checking candidate yaws, grasp heights, and finger pre-shapes with inverse kinematics and collision detection on a scratch copy of the state, approaches with a trapezoidal velocity profile (1.6 m/s$^2$), closes and verifies a two-finger grasp (up to three attempts), carries $o^\star$ at reduced speed and acceleration (0.24 m/s, 0.8 m/s$^2$) above every surface on the straight path to the goal while correcting for its offset in the hand, rotates the hand if the fingers would collide at the goal, lowers and re-centers $o^\star$ until it is supported, lifts and re-centers after a stalled insertion (up to three times), waits for $o^\star$ to come to rest, and opens until neither finger touches it before retracting. The oracle uses state that policies do not observe, so its success bounds solvability rather than predicting learned performance.

### B. Demonstration Scale and Packaging

The generator records 50 successful oracle demonstrations per task (10,500 in total) in Robomimic HDF5 with 90%/10% train/validation masks and per-episode metadata. Each demonstration starts from its own seed, so the 50 demonstrations of a task start from distinct layouts. Successful oracle episodes with the controller of this paper average 108 steps, so a release of this size holds about $1.1 \times 10^6$ timesteps; releases made with the kinematic grasps of earlier versions must not be mixed with

TABLE I

**TAXONOMY OF THE AFFORD-BENCH SUITE.** 210 TASKS IN 7 DOMAINS. TOLERANCES ARE THE SUCCESS THRESHOLDS $(\epsilon_{xy}, \epsilon_z, \epsilon_\theta)$; D5 TASKS TAKE THE TIER OF THEIR SCENE. D1, D3, AND D7 SCENES ADD TWO DISTRACTOR BODIES; TOLERANCE-INDEXED VARIANTS ARE A STAGE 2 DELIVERABLE.

| Domain | Task IDs | Count | Representative Manipulation Tasks | Evaluated Bodies | Tol. (mm / deg) |
|---|---|---|---|---|---|
| **D1: Industrial Precision** | 001–030 | 30 | Shaft-in-bore, gear-on-shaft, pin-in-hole, kitting, fixture contact | Shaft, spur gear, dowel pin | 2 / 5 |
| **D2: Culinary Operations** | 031–060 | 30 | Burger layer stacking, mug to drip tray, capsule to hopper | Patty, cheese, tomato, buns, mug, capsule | 8 / 15 |
| **D3: Domestic Tabletop** | 061–090 | 30 | Color sorting, block stacking, relocation, touch | Two cubes, one cylinder | 8 / 15 |
| **D4: Laboratory Sciences** | 091–120 | 30 | Tube racking, centrifuge loading, pipette positioning | Conical tube, microtube, pipette | 2 / 5 |
| **D5: Long-Horizon Sequences** | 121–150 | 30 | Multi-step assembly, sorting, and service instructions | One per task (Stage 1) | 2 or 8 / 5 or 15 |
| **D6: Precision Chaining** | 151–180 | 30 | 2–3 step industrial and laboratory chains | One per task (Stage 1) | 2.5 / 5 |
| **D7: Service Assembly** | 181–210 | 30 | 2–3 step kitchen and household chains | One per task (Stage 1) | 8 / 15 |
| **Total** | **001–210** | **210** | **7 Domains × 30 Tasks (35 Evaluation Suites)** | **16 rigid bodies** | **2–8 / 5–15** |

it. The converters of Sec. VIII export it to RLDS (OpenVLA, Octo) and LeRobot (SmolVLA).

## VI. TRAINING AND EVALUATION PROTOCOLS

### A. Standardized Closed-Loop Rollout Procedure

Each policy step applies one action at 20 Hz: the arm is interpolated to the IK joint target over 25 physics steps of 2 ms, with commanded TCP speed capped at 0.5 m/s. For each task we roll out $N = 50$ episodes with evaluation seeds disjoint from data-generation seeds, with $H = 400$ steps for D1–D4 and $H = 800$ for D5–D7. Every suite, including nominal, draws its initial state from the evaluation seed.

### B. Standardized Evaluation Suites

Models are evaluated on 35 suites (7 domains × 5 conditions): (1) **Nominal**; (2) **Spatial** ($\mathcal{P}_{\text{spatial}}$); (3) **Visual** ($\mathcal{P}_{\text{visual}}$); (4) **Physical** ($\mathcal{P}_{\text{physics}}$); and (5) **Composite** ($\mathcal{P}_{\text{comp}}$), with parameters as in Sec. IV. Language variants are evaluated as a separate axis on the nominal scene.

### C. Quantitative Evaluation Metrics

We report: (i) **Success Rate** $\text{SR} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}[\exists\, t \le H : s_t^{(i)} \in \mathcal{C}_{\text{aff}}]$ with Wilson 95% intervals over episodes and mean ± s.d. over 3 training seeds; (ii) **Physical Robustness Index** $\text{PRI}(\mathcal{P}) = \text{SR}(\mathcal{P})/\text{SR}(\text{Nominal}) \times 100\%$, defined when $\text{SR}(\text{Nominal}) > 0$; (iii) **Generalization Degradation Index** $\text{GDI} = 100\% - \text{PRI}$; and (iv) **Force Violation Rate** (FVR), the fraction of episodes violating $F_{\max}$.

## VII. SCIENTIFIC HYPOTHESES & PRELIMINARY CALIBRATION

### A. Core Scientific Hypotheses

AFFORD-Bench is designed to test three hypotheses; $H_2$ requires tolerance-indexed variants and $H_3$ multi-body predicates (both Stage 2):

- $H_1$ **(physical versus appearance shift)**: *policies with high nominal success lose more under $\mathcal{P}_{\text{physics}}$ than under $\mathcal{P}_{\text{visual}}$ of matched severity, with object 2D coordinates unchanged.*
- $H_2$ **(tolerance fragility)**: *at fixed spatial shift, success decreases monotonically as $\epsilon_{xy}$ shrinks; tested with tolerance-indexed variants of D1 and D4 tasks at $\epsilon_{xy} \in \{2, 4, 8, 16\}$ mm.*

TABLE II

**STAGE 1 CALIBRATION.** SUCCESS RATE (%) ON THE NOMINAL SUITE (L1) AND, FOR THE ORACLE, THE PHYSICAL SUITE (L4). ORACLE: PRIVILEGED CLOSED-LOOP EXPERT, 5 RESET SEEDS PER TASK (150 EPISODES PER DOMAIN). RANDOM: UNIFORM ACTIONS, 5 EPISODES PER TASK. NULL: NEVER MOVES, GRIPPER OPEN, 1 EPISODE PER TASK. RANDOM AND NULL USE $H = 400$; THE ORACLE USES AT MOST 600 STEPS. NO TASK SUCCEEDS AT RESET. BC AND DIFFUSION POLICY RESULTS ARE PENDING.

| Task Domain | Oracle | Oracle (L4) | Random | Null | BC | Diffusion |
|---|---|---|---|---|---|---|
| D1: Industrial Precision | 100.0 | 37.3 | 0.0 | 0.0 | TBD | TBD |
| D2: Culinary Operations | 100.0 | 56.7 | 0.0 | 0.0 | – | – |
| D3: Domestic Tabletop | 100.0 | 41.3 | 0.0 | 0.0 | TBD | TBD |
| D4: Laboratory Sciences | 100.0 | 84.0 | 0.0 | 0.0 | – | – |
| D5: Long-Horizon Sequences | 100.0 | 38.0 | 0.0 | 0.0 | – | – |
| D6: Precision Chaining | 99.3 | 51.3 | 0.0 | 0.0 | – | – |
| D7: Service Assembly | 99.3 | 22.7 | 0.0 | 0.0 | – | – |
| **All 210 tasks** | **99.8** | **47.3** | **0.0** | **0.0** | **Stage 2** | |

- $H_3$ **(error compounding)**: *in D6 and D7, the success probability of a $k$-step chain falls below the product of its per-step success rates measured in isolation.*

### B. Preliminary Empirical Calibration

Table II reports solvability under the protocol of Secs. III and IV. The oracle solves 1048 of 1050 nominal episodes (99.8%; episode-level Wilson 95% interval 99.3–99.9%, which ignores correlation between episodes of the same task) and 208 of 210 tasks in every seed. One episode each of tasks 168 and 203 fails; in task 168 the released pin settles 3.6 mm from its goal against a 2.5 mm tolerance. Most of the earlier losses came from bodies slipping or being knocked out of the fingers in transport, which slower carrying above every surface on the path removed. Random actions solve none of 1050 episodes (upper 95% bound 0.4%) and the null policy none of 210 (1.8%); the lift requirement and the release-and-rest predicate remove trivially satisfied goals. Under L4 the same oracle, which does not adapt its grip, succeeds in 47.3% of episodes (44.3–50.4%), from 84.0% in D4 to 22.7% in D7; we have not established what fraction of L4 episodes any policy can solve. In 8 of 630 resets (1.3%) no valid scattered start existed and the nominal pose was used; task 12, a gear seated on its shaft, always starts from it. All results use one simulator build (MuJoCo 3.2.7) and MuJoCo Menagerie meshes for the Panda. Learned baselines on D1 and D3 will use 3 training seeds and 50 evaluation episodes per task.

## VIII. Discussion and Stage 2 Extension Charter Roadmap

### A. Physical Hardware Correspondence and Sim-to-Real Feasibility

Controlled comparison across 210 tasks and 35 suites requires what hardware cannot provide: exact resets, perturbations sampled from known distributions, and ground-truth success labels. The full matrix of 50 episodes per task and condition is 52,500 rollouts per policy, each replayable from its seed; at two physical trials per minute including resets, that is about 440 robot-hours per policy, over which wear, recalibration, and lighting drift would confound the comparison. Simulation also exposes the quantities the perturbations act on, such as object pose, friction, and mass, so success is a measured predicate rather than an annotator judgment. We therefore treat simulation as the instrument for comparing policies and use hardware to test whether the rankings it produces hold on physical robots.

The observation and data interfaces mirror common real-robot pipelines but are not identical to them. Each episode records a fixed third-person RGB stream, a wrist RGB stream, optional workcell metric depth, and joint and TCP proprioception, the same modality set as Franka workcells instrumented with Intel RealSense cameras and as ALOHA-style [27] and LeRobot [4] setups with fixed and wrist cameras. Episodes are stored in Robomimic HDF5, and the released converters `scripts/convert_to_lerobot.py` and `scripts/convert_to_rlds.py` translate them losslessly into LeRobot v2.0 datasets [4] and the RLDS episode layout of Open X-Embodiment [16] (verified by round-trip tests), so one training pipeline can consume simulated and physical episodes. Three gaps remain: the simulated image sizes ($640 \times 660$, $300 \times 420$) differ from native RealSense modes and must be matched by cropping or re-rendering; the action is a Cartesian position increment, whereas SO-101 and Piper arms are commanded in joint space and ALOHA in bimanual joint space; and simulated bodies and fingertips are rigid, with contact softness set by the solver rather than by measured compliance. The pilot below is designed around these gaps.

The Stage 2 Extension Charter includes a hardware pilot on two D1 tasks: peg insertion (Task 003) and gear meshing onto a seated shaft (Task 002). The primary platform is a Franka Panda with wrist and fixed RealSense cameras, Cartesian impedance control, and 3D-printed replicas of the simulated jig, pin, shaft, and gear at three clearances. Policies trained on simulated demonstrations, physical teleoperated demonstrations, and their union are evaluated under nominal, spatial ($\pm 3$ cm, $\pm 45°$), and physical (surface films for friction, added mass) conditions that replicate L2 and L4. We report the embodiment, 20 trials per condition, success as full seating verified by a depth gauge, and a fixture-based reset protocol, and we measure sim-to-real agreement as the rank correlation of per-condition success between simulation and hardware. SO-101 and Piper arms repeat the pilot at the loosest clearance as a low-cost cross-embodiment check rather than a precision test. The pilot is positioned against IndustReal [23], which transfers RL insertion and gear-meshing policies on a Franka: our question is whether simulated perturbation rankings predict physical degradation for imitation and VLA policies, not whether a single policy transfers.

### B. Stage 2 Extension Charter

In accordance with the RSS 2027 shepherding process, the Stage 2 charter comprises: (1) *Simulator fidelity*: tolerance-indexed task variants, multi-body predicates for D5–D7, and compliant fingertip models; (2) *Full empirical scaling*: 50 rollouts per condition $\times$ 210 tasks $\times$ 35 suites for Diffusion Policy, ACT, OpenVLA, SmolVLA, AtomVLA, $\pi_0$, SpatialVLA, CoT-VLA, and DP3; (3) *Disentangled ablations*: separate sensitivity curves for $\mathcal{P}_{\text{spatial}}$, $\mathcal{P}_{\text{physics}}$, $\mathcal{P}_{\text{visual}}$, and tolerance; (4) *Hardware pilot*: peg insertion and gear meshing as described above; and (5) *Domain 8 (Tasks 211–240)*: tool-mediated manipulation (hook retrieval beyond the workspace, spatula wedging, hex-key socketing), non-prehensile manipulation (edge singulation, fixture-assisted pivoting, corral sweeping), twist-lock and detent assembly (bayonet, key and latch, push-and-turn caps), and mid-trajectory reactive perturbations.

Pick-and-place suites reward reaching a pose; Domain 8 tasks succeed only through a mechanism the policy must infer, such as a tool frame, a pushing motion cone, a configuration-space slot, or a displaced target, so they test physical affordance reasoning in VLAs and world models more directly. Each task enters the benchmark only if a privileged oracle solves at least 95% of its episodes and null and random policies solve none; tasks that need wrist roll and pitch, which Stage 1 does not execute, are marked as such. The perturbation test is already validated on Stage 1 tasks (`scripts/test_reactive_perturbation.py`): with a 4 cm displacement at mid-approach, a closed-loop and a 16-step chunked execution of the same state-feedback law succeed in 18 of 19 episodes, open-loop replay in 0 of 19, and all three in 9 of 9 without the displacement. Specifications are in `docs/FRONTIER_COMPLEX_TASKS.md`.

## REFERENCES


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